

Chapter 2

Medical Terminology for Central Service Technicians

Learning Objectives

As a result of successfully completing this chapter, readers will be able to:

1. Explain the importance of medical terminology for Central Service technicians
2. Identify the various elements used in medical terminology including prefixes, roots and suffixes
3. Discuss how medical terminology can refer to the human anatomy, disease processes, surgical instruments and surgical procedures
4. Understand medical terminology used to refer to surgical procedures in surgery schedules
5. Understand the importance of medical terminology for service quality in the Operating Room

INTRODUCTION

Central Service (CS) technicians require knowledge of medical terms to help them succeed on the job. The special terminology used to describe parts of the body, diseases, instruments and surgical procedures will be addressed in this chapter.

IMPORTANCE OF MEDICAL TERMINOLOGY

The Healthcare Profession

The Association of periOperative Registered Nurses (AORN) specifically states that "skilled and competent allied health care providers and support personnel are valued members of the perioperative care team, contributing to safe patient care and positive patient outcomes." As such, CS technicians must have a grasp of the medical terminology that is used by healthcare customers, as well as by fellow CS professionals. The American National Standards Institute (ANSI) and the Association for the Advancement of Medical Instrumentation (AAMI), in ANSI/AAMI ST79, 4.2.2, specifically states that CS technicians must be "knowledgeable and competent" to adequately perform the vital tasks that they perform. Indeed, The Joint Commission (TJC) states that in order to maintain a reliable system for medical device processing, institutions must place an emphasis upon the "orientation, training and competency of health care workers" responsible for this task. Understanding medical terminology is part of the training and competency that CS technicians should possess.

Understanding the Operating Room

CS professionals must provide Operating Room (OR) personnel with the instruments and supplies needed for surgical procedures. As these needs are conveyed, CS technicians must also understand the terminology/language spoken to them in order for them to provide quality customer service and contribute to positive patient outcomes.

For example, if the OR calls for instrumentation needed for an emergency pericardial window, a knowledgeable CS technician will know that this is a cardiac procedure that involves cutting into the pericardium (a membranous sac that surrounds the heart) in order to drain fluid from the pericardial space into the pleural cavity. Also, if the OR calls for a case cart for a **STAT** Repair of an Abdominal Aortic Aneurysm (AAA), this request will be responded to immediately as the CS technician will understand that the patient is at risk of losing his/her life. If the OR issues a request for a case cart for a routine Laparoscopic Cholecystectomy, the CS technician will know that this is not an emergency case. The case cart will still be picked and the instruments and supplies will still be prepared and organized; however, this is usually not an emergency situation.

STAT Abbreviation for the Latin "statim" meaning immediately or at once.

Providing Quality of Service to the Patient

It is through knowing and understanding medical terminology that CS technicians are able to understand what is being asked of them. This knowledge of medical terminology enables the CS technician to react in an appropriate manner when the OR or medical staff makes a request; however, the opposite is also true. If the OR makes a request that is not understood, the productivity of the OR is compromised. This interruption of the perioperative process will compromise the quality of service rendered to the patient and may cause a delay in treatment. Nothing is more important than quality. For this reason, TJC and the Centers for Disease Control and Prevention (CDC) recommend that all healthcare institutions provide comprehensive and intensive training for all staff involved in the processing of medical instrumentation.

ANATOMY OF A MEDICAL TERM

CS technicians frequently encounter specialized medical terms in their daily work activities, and each of those terms contains key elements to help staff better understand the words' origin and meaning. As technicians learn the meanings of these names, they will have a greater understanding of what their work involves, and their competency and job satisfaction will increase.

Word Elements

The majority of medical terms are of either Greek or Latin origin. The word "pericardium," for example, is composed of two Greek **word elements**: peri (meaning around), and kardia (meaning heart). Hence, the term is used to refer to the membranous sac that surrounds the heart, as well as the roots of the great vessels of the heart. The term "rigor mortis" is composed of two Latin word elements: rigor (stiffness) and mortis (meaning "of death"); hence, the term is used to refer to the stiffening of the body that occurs after death. Many terms combine both Greek and Latin word elements to form a single medical term. For example, the term "claustrophobia" (meaning fear of enclosed spaces) joins the Latin word element "claustrum" (enclosed space) to a Greek word element "phobia" (fear). Another example of this blending of Greek and Latin in the formation of medical terms can be seen in the term "autoclave" (a term that is very important for CS). The term "auto" (Greek) refers to self while the term "clavis" (Latin) refers to a key; hence, an autoclave is a type of self-locking device (a sterilizer is designed to not open until its cycle is finished).

At first, medical terminology may seem daunting and overwhelming; however, after a working knowledge of these word elements is gained, it becomes easier to analyze and use the words effectively.

Medical terminology changes with the dynamics of evolving technology in health

(intraoperative), and after (postoperative). The prefix “peri” is also found in the surgical instrument called periosteal elevator. Here, the prefix “peri” (around) is attached to the root “osteo” (bone); hence, a periosteal elevator is an instrument used to remove tissue from around a bone (the periosteum). Even the term “abnormal” functions in this way. The “ab” functions as a prefix to negate the meaning of the root “normal”; hence, the meaning of the word is not “normal.”

- **Suffix word element** Comes after the root. When added to the root (at the end of a word), the suffix can also alter or modify its meaning. For example, the medical term for the suffix that means inflammation is “itis”; therefore, the term “pericarditis” refers to inflammation (itis) around (peri) the heart (cardio). Most of us are acquainted with the term “tonsillitis” as referring to the inflammation of the tonsils. Bronchitis refers to the inflammation of the bronchi, which are the tubes extending from the trachea into both sides of the lungs; hence, whenever the suffix “itis” is found at the end of a medical term (modifying a root word preceding it), a meaning of inflammation will be present.

Of special note with the term “pericarditis” is that the “o” in the root “cardio” is dropped in the word “pericarditis.” This reflects something called a **combining vowel**. In many cases, a combining vowel (usually an “o”) is either added to a root or removed from it to ease the pronunciation of the word. The use of combining vowels can result in some rather long medical words. The linguistic effect of this addition or elimination of the combining vowel is to ease the pronunciation of the word. For example, the term “herniorraphy” which refers to the suturing/repairing (-rrhaphy) of a rupture/hernia (herni), contains an “o” which has been added as a combining vowel (herni-o-rrhaphy). Conversely, the term “proctitis,” which refers to inflammation of the rectum, features the dropping of the combining vowel “o” (procto/rectum

+ itis/inflammation = proctitis). While the addition of a combining vowel is common in medical terminology, it is important to understand that not all medical terms have combining vowels.

Word elements Parts of a word.
Root word element Tells the primary meaning of a word; also called base word element.
Prefix word element The word element that comes before the root word element.
Suffix word element The word element that comes after the root word element.
Combining vowel A letter, usually an “o”) that is sometimes used to ease the pronunciation of a medical word.

An easy way to remember the difference between prefix (which comes before the root) and suffix (which comes after the root) is to put the words in alphabetical order: prefix, root and suffix. This tells you that prefix is the first word element and suffix is the last word element.

Note: Not all medical terms consist of all three word elements. A medical term may be formed by a root alone, by combining two roots, a root and suffix, or a prefix and root. Some medical terms can even be formed by combining three roots together.

The best way to learn the meaning of a medical term is to analyze and understand its components, so that the word can be taken apart. Begin with the suffix (if present) since it most often gives a clue and meaning about the root, and how it is being used. Then consider the root and prefix (if present). In other words, consider the overall relationship between each word element in the term. Medical terms can be a lot like building blocks; if one can figure out how they fit together, one can determine what they mean.

Prefix	Root	Suffix	Word
hemi- (half)	arthro (joint)	-plasty (surgical restoration)	hemiarthroplasty (surgical restoration of half of a joint, the femoral portion of the hip joint, the proximal femur, a form of hip replacement surgery)
hemi- (half)	gastro (stomach)	-ectomy (surgical removal)	hemigastrectomy (removal of half of the stomach) <i>Note: The combining vowel "o" is dropped in this word.</i>
hemi- (half)	colo (colon)	-ectomy (surgical removal)	hemicolectomy (removal of half of the large intestine) <i>Note: The combining vowel "o" dropped from this word.</i>
para- (beside, near)	thyroid (thyroid)	-ectomy (surgical removal)	parathyroidectomy (surgical removal of parathyroid glands)
septo- (dividing wall)	rhino (nose)	-plasty (surgical restoration)	septorhinoplasty (surgical restoration of the nose) <i>Note: Two roots combined with a suffix.</i>
chole- (bile)	cyst (fluid-filled sac)	-ectomy (surgical removal)	cholecystectomy (surgical removal of the gallbladder) <i>Note: Two roots combined with a suffix.</i>
electro- (electrical activity)	cardio (heart)	-gram (written record of)	electrocardiogram (written record of the electrical activity of the heart) <i>Note: Two roots combined with a suffix.</i>
electro- (electrical activity)	encephalo (brain)	-gram (written record of)	electroencephalogram (EEG) (the tracing of brain wave activity) <i>Note: Two roots combined with a suffix.</i>

Figure 2.1 Combined Word Elements

Figure 2.1 shows additional examples of how word elements are combined to form medical terms. Sample words shown here contain a prefix, a root and a suffix. While not all medical terms feature all three word elements in the same word, it is important to understand that many medical terms do feature all three word elements.

When analyzing medical terms, several suffixes meaning “pertaining to” may be encountered. Figure 2.2 provides several examples of suffixes, words in which they are used, and the meaning of the words.

Figure 2.2 Suffixes that Mean “Pertaining To”

Suffix	Example	Meaning
-ac	cardi-ac	pertaining to the heart (“o” in cardio is dropped)
-al	derm-al	pertaining to the skin
-ic	hem-ic	pertaining to blood
-eal	esophag-eal	pertaining to the esophagus
-ary	pulmon-ary	pertaining to lungs
-ous	cancer-ous	pertaining to cancer

Just as with the English language, certain rules of grammar apply to medical terminology. Making medical terms conform to the basic rules of spelling and pronunciation may result in letters in the word element being changed, dropped or added. **Figure 2.3** shows some examples.

Figure 2.3 Letters in Word Elements May Be Dropped or Added

Prefix	Root	Suffix	Word	Letters Changed
	procto (rectum)	-itis (inflammation)	proctitis	“o” is dropped
	broncho (bronchus)	-itis (inflammation)	bronchitis	“o” is dropped
	endo (within)	-oscopy	endoscopy	“o” is dropped
	artery (artery)	-ectomy (removal)	endarterectomy	“o” and “y” are dropped
	fascia	-otomy	fasciotomy	“a” is dropped
	chir (hand)	-plasty (surgical repair)	chiroplasty	“o” is added
	herni (rupture)	-rrhaphy (to suture)	herniorrhaphy	“o” is added
	hyster (uterus)	-pexy	hysteropexy	“o” is added

As noted above, when analyzing a medical term, it is sometimes best to start with the meaning of the suffix. In this way, one can readily identify if the word relates to a surgical procedure, a medical condition or a portion of the human anatomy. **Figures 2.4** and **2.5** list some common medical and surgical suffixes. Suffixes are word elements that come after the root word, and may be one or two syllables.

Figure 2.4 Common Medical Suffixes

Suffix	Meaning	Example	Combined Meaning
-algia	pain	neuralgia	nerve pain
-cide	kill	bactericide	a substance that kills bacteria
		virucide	a substance that kills viruses
		fungicide	a substance than kills fungi
-emia	blood	hyperglycemia	high blood sugar
		hypoglycemia	low blood sugar
-genic	origin	osteogenic	originating in the bones
		iatrogenic	an adverse effect or complication originating from a physician
-gram	record/image of	mammogram	a radiographic image of the breast for early detection of breast cancer
		cholangiogram	a radiographic image of the bile ducts using contrast medium to check for blockage (frequently done during a cholecystectomy)
		arthrogram	a radiographic image of a joint after injection of a contrast medium
		angiogram	a radiographic image of the inside, or lumen, of blood vessels and organs of the body using a contrast agent
-itis	inflammation	tonsillitis	inflammation of the tonsils
		hepatitis	inflammation of the liver
		arthritis	inflammation of a joint
		bronchitis	inflammation of the bronchi
		meningitis	inflammation of the meninges (the membranous layer surrounding the brain and spinal cord)
-megaly	large or enlargement	cardiomegaly	enlargement of the heart
-necrosis	death of tissue	arterionecrosis	tissue death of an artery
-ology	study of	bacteriology	the study of bacteria
		oncology	study of cancer
		neurology	study of the nervous system
		cardiology	study of the heart
		nephrology	study of the kidney
-oma	tumor	carcinoma	malignant tumor
		myoma	tumor consisting of muscular tissue
		meningioma	tumor of the meninges (the membranous layer surrounding the brain and spinal cord)
		fibroadenoma	a breast lump composed of fibrous and glandular tissue
		papilloma	a benign tumor which grows on the skin or mucous membrane, can be caused by a virus
-pathy	disease	encephalopathy	disorder or disease of the brain
		cardiomyopathy	disorder or disease of the heart
-rrhage	flow	hemorrhage	uncontrolled flow of blood

Figure 2.5 Common Surgical Suffixes

Suffix	Meaning	Example	Combined Meaning
-cise	cut	excise	to cut out
		incise	to cut into
-ectomy	surgical removal	cystectomy	removal of a cyst
		tonsillectomy	removal of the tonsils
		pneumonectomy	removal of a lung
		laminectomy	removal of a portion of a lamina (part of the vertebra in the spine)
		hysterectomy	removal of the uterus
		appendectomy	removal of the appendix
		thrombectomy	removal of a blood clot
		orchiectomy	removal of a testicle
		nephrectomy	removal of a kidney
		vitrectomy	removal of some or all of the vitreous humor from the eye
-oscopy	visual examination and possible treatment of an organ or joint	hemorrhoidectomy	removal of swollen or inflamed vascular structures in the anal canal
		bumionectomy	removal (realignment) of a misaligned bone in the big toe
		discectomy	removal of a herniated disc in the spine
		microdiscectomy	minimally invasive removal of a herniated disc in the spine
		thyroidectomy	removal of the thyroid gland
		laparoscopy	visual examination of organs in the abdomen
		arthroscopy	visual examination of a joint
		cystoscopy	visual examination of the bladder
		bronchoscopy	visual examination of the bronchi
		colonoscopy	visual examination of the large intestine
-ostomy	creation of an opening	thoracoscopy	visual examination of the thoracic cavity
		fluoroscopy	an imaging technique which provides live images (motion included) of a surgical site during surgery, referred to as C-Arm due to its shape
		colostomy	creation of new opening to colon
		tracheostomy	creation of new opening to trachea
		urostomy	creation of new opening for the urinary system
		nephrostomy	creation of a new opening to the kidney for the urinary system

Suffix	Meaning	Example	Combined Meaning
		ventriculostomy	creation of an opening within a cerebral ventricle for drainage
		ileostomy	creation of a new opening to the ileum (a portion of the small intestine near the colon/large intestine)
-otomy	incision into an organ	craniotomy	incision into the skull
		thoracotomy	incision into the pleural space of the chest
		arthrotomy	incision into a joint
		osteotomy	incision into a bone
		fasciotomy	incision into the fibrous membrane (fascia) which covers a muscle
		urethrotomy	incision into the urethra
		fistulotomy	incision into a fistula (an abnormal connection between two organs or vessels)
		sternotomy	incision into/splitting of the sternum (breastbone)
-pexy	surgical fixation	orchiopexy	surgical fixation of an undescended testicle to the correct location
		hysteropexy	abdominal fixation of the uterus
-plasty	surgical restoration	rhinoplasty	surgical repair of the nose
		arthroplasty	surgical repair of a joint
		cranioplasty	surgical repair of the skull
		tympanoplasty	surgical repair of the ear drum
		urethroplasty	surgical repair of the urethra
		kyphoplasty	surgical repair of a fractured vertebra by percutaneous (through the skin) injection of bone cement
-rrhaphy	to suture	myorrhaphy	to suture a muscle wound
		herniorrhaphy	to suture a hernia
-tome	a cutting instrument	dermatome	an instrument used for cutting skin
		osteotome	an instrument used for cutting bone

As can be seen by the brief listing of suffixes in **Fig res 2.4** and **2.5**, medical terms can be complex and can mean entirely different things depending on how they are composed. For example, while the word “tonsillitis” means inflammation of the tonsils, the word “tonsillectomy” refers to the surgical removal of the tonsils. Herein lies the adventure of building a medical terminology vocabulary: words can mean entirely different things depending on how the root is used in concert with a prefix, suffix or both. **Fig res 2.6** and **2.7** contain a listing of common roots found in medical terminology. Roots are base word elements that refer to the main body of a medical word.

Figure 2.6 Common Medical Roots

Root	Meaning	Example	Meaning
adeno	gland	adenoma	glandular tumor
aero	air	aerobic	requiring oxygen for growth
		anaerobic	not requiring oxygen for growth
arthro	joint	arthritis	inflammation of a joint
broncho	bronchus	bronchoscope	endoscope used to visualize the bronchi
cardio	heart	endocarditis	inflammation of the inner layer (endocardium) of the heart
		cardiomyopathy	heart muscle disease
		cardiopulmonary resuscitation (CPR)	emergency efforts to revive a person in cardiac arrest by means of chest compressions and rescue breaths
		myocardium	cardiac/heart muscle
cerebro	brain	cerebrospinal	referring to the brain and spinal cord
chole	bile	cholecyst	gallbladder
chondro/io	cartilage	chondroma	cartilaginous tumor
costo	rib	intercosto	between the ribs
cysto	bladder	cyst	any fluid-filled sac
cyto	cell	erythrocyte	red blood cell
derma	skin	dermopathy	skin disease
gastro	stomach	gastrointestinal	pertaining to the stomach and intestines
gyne	woman	gynecology	the study of diseases affecting the female
hema or hemat	blood	hemophilia	inability of the blood to clot
		hemostat	instrument used to control bleeding
		hematoma	collection of blood in tissue
		hemodialysis	filtration of the blood mechanically, occurring outside the body
hepat	liver	hepatitis	inflammation of the liver
leuko	white	leukocyte	white blood cell
		leukemia	a form of cancer of the blood or bone marrow involving abnormal white blood cells
nephros	kidney	nephritis	inflammation of the nephrons in the kidneys
		nephrosis	degenerative disease of the kidneys
thrombus	blood clot	thrombosis	formation of a blood clot inside a blood vessel obstructing the flow of blood (DVT= deep vein thrombosis)

Figure 2.7 Common Surgical Roots

Root	Meaning	Example	Meaning
arthro	joint	arthroscopy	visual examination of a joint
		arthrocentesis	joint puncture for the aspiration of synovial fluid
		arthrodesis	surgical fusion of a joint
colo	colon	colectomy	removal of part of the large intestine
cranio	skull	craniotomy	surgical opening into the skull
herni	rupture	herniorrhaphy	surgical repair of a rupture
hyster	uterus	hysteropexy	abdominal fixation of the uterus
		hysteroscopy	visual examination of the uterus
lipo	fat	liposuction	aspiration of fat cells
litho	stone	lithotripsy	crushing of a stone
mast	breast	mastectomy	surgical removal of a breast
oopher	ovary	oophorectomy	surgical removal of an ovary
osteo	bone	osteosynthesis	surgical reduction and fixation of a bone fracture
rhino	nose	rhinoplasty	surgical repair of the nose
tracheo	trachea	tracheostomy	creation of new opening to trachea
stoma	opening	anastomosis	a connection or reconstruction of two separate tubular structures; for example, blood vessels or portions of intestines

Figures 2.6 and 2.7 above list common roots that show how medical terms can mean a variety of things depending on how they are constructed. Roots can be modified by prefixes, suffixes or both. Roots can also be combined to form their own words, or they can be modified by prefixes/suffixes. Meanwhile, the terms themselves can refer to either a portion of the human anatomy, a disease process or a surgical procedure. While this might seem difficult at first, it becomes much easier when one learns how to identify the suffixes and prefixes (if present) attached to the root/roots involved. By learning common suffixes and prefixes, the reader can identify whether the term is referring to the human anatomy, a disease process, a surgical procedure or a surgical instrument. **Figures 2.8 and 2.9** provides a listing of common prefixes used in medical terminology. Prefixes are word elements that are placed before the root to alter or modify its meaning.

Figure 2.8 Common Medical Prefixes

Prefix	Meaning	Example	Combined Meaning
a, an-	without	asepsis	without infection; sterile
		anesthesia	without sensation (local or general)
		atraumatic	not inflicting injury or wound
		analgesia	without pain
		atrophy	a reduction in size of a part of the body (wasting away) due to poor circulation, poor nutrition or a disease process.
ad-	toward (in the direction of)	addiction	toward dependence on a drug
ante-	before	antepartum	before the onset of labor
anti-	against	antiseptic	preventing sepsis (infection)
dis-	apart, away	hip dislocation	displacement of femur from pelvic joint
		disinfectant	chemical used to kill microorganisms
dys-	painful	dysentery	painful inflammation of the intestine
endo-	within	endotracheal	within the trachea
		endoscope	instrument used to visualize a joint or organ in the body
		endoscopy	visualization within, and possible treatment of, a part of the body using an endoscope and endoscopic instrumentation
extra-	outside	extracorporeal	outside of the body
hyper-	above, excessive	hyperacidity	excessive acid in the stomach
		hypertensive	high blood pressure
		hypertrophy	excessive growth/size of a part of the body due to cellular enlargement
hypo-	below, deficient	hypoglycemia	low sugar content in the blood
		hypotensive	low blood pressure
inter-	between	intercellular	between, or among, cells
		interstitial	an empty space
		interdepartmental	between departments
		intercostal	between the ribs
intra-	within, inside	intravenous	in, or into, a vein
		intramuscular	located in, or injected into, a muscle
		intraabdominal	within the abdomen
		intraoperative	during surgery
		intraarticular	within a joint
		intraocular	within the eye
		intracranial	within the skull
neo-	new	neonatal	newborn
per-	through	percutaneous	through the skin
post-	after	postpartum	after delivery of a baby
		postoperative	after surgery
pre-	before	preoperative	before surgery
sub-	under, beneath	subcutaneous	beneath the skin
		subclavian	located under the clavicle
		subdural	located under the dura mater (the outermost membrane surrounding the brain)
supra-	above	suprapubic	above the pubis
		supracondylar fracture	fracture of the distal humerus above the elbow joint

Figure 2.9 Common Surgical Prefixes

Prefix	Meaning	Example	Meaning
bi-	two/both sides	bilateral total hip reconstruction (THR)	two (both) total hip reconstructions
		bilateral myringotomy with tubes	incision into (both) ear drums (tympanic membrane) in order to place drainage tubes
		bilateral salpingo-oophorectomy	surgical removal of both fallopian tubes and ovaries
		bilateral strabismus repair	surgery performed on the extraocular muscles to correct eye misalignment
hemi-	half	hemigastrectomy	surgical removal of half of the stomach
		hemiarthroplasty	a form of hip replacement surgery
		hemicolectomy	removal of half of the large intestine
para-	beside, near	parathyroidectomy	surgical removal of parathyroid glands
		paratracheal	beside the trachea
peri-	around, about	periosteal elevator	instrument used to remove tissue around the bone (the periosteum)
		perioperative	all aspects of the surgical process (before, during and after)
		pericardium	the serous (fluid emitting) membrane that stretches around the heart and lines the mediastinum
		peritoneum	the serous (fluid emitting) membrane that stretches around the abdominal cavity and is its lining
post-	after	postoperative	after surgery
trans-	across, through	transanal	through the anus
		transoral	through the mouth
		transesophageal	through the esophagus
		transurethral	through the urethra

Understanding Usage

Learning medical terminology can be an almost endless process, which multiplies as various body parts, procedure types and disease processes are combined to mean different things. As new surgical procedures are created (utilizing new approaches, instruments and technologies), new medical terminology will also be created to refer to these

procedures. CS technicians must understand the language spoken to them by the OR and medical staff in order to provide them with the goods and services they need for their procedures. Language frequently used by the OR utilizes abbreviations or acronyms to refer to various types of surgeries. **Figure 2.10** contains a brief listing of some surgical procedure abbreviations or acronyms.

Figure 2.10 Abbreviations/Acronyms for Surgical Procedures

Abbreviation	Surgical Procedure	Meaning
AAA	Repair of abdominal aortic aneurysm	Surgical repair of a weakening/ballooning area of the abdominal portion of the aorta (the largest artery in the body).
ACL	Anterior cruciate ligament	Reconstruction or repairing of the anterior cruciate ligament. In an ACL reconstruction a graft is used to replace the ligament, and in an ACL repair the torn ligament is put back together.
ACF	Anterior cervical fusion	Surgical fusion of vertebrae in the cervical spine, approached from the front side of the patient's body.
ALIF	Anterior lumbar interbody fusion	Spinal surgery in which bone grafts/implants are used to fuse vertebrae in the lumbar spine, approached from the front side of the patient's body.
AKA	Above the knee amputation	Surgical removal of the leg above the knee.
AV Graft	Arteriovenous graft	A surgical connection is made between an artery and a vein to allow hemodialysis access.
BMT	Bilateral myringotomy with tubes	Incision into the eardrum (tympanic membrane, both sides) for drainage via tube placement.
BSO	Bilateral salpingo-oophorectomy	Surgical removal of both fallopian tubes and ovaries.
BKA	Below the knee amputation	Surgical removal of the leg below the knee.
CABG	Coronary artery bypass graft	Creation of a new blood supply to an area of the heart with a clogged/blocked artery using the patient's own blood vessel to function as the graft (frequently harvested from the leg).
CR	Closed reduction	Treatment of a fractured bone without a surgical incision.
D&C	Dilation and curettage	Dilation of the uterine cervix and scraping of the inner lining of the uterus (endometrium) with a uterine curette.
EGD	Esophagogastroduodenoscopy	Endoscopic procedure that visualizes the upper portion of the gastrointestinal tract up to the duodenum (where the stomach connects to the small intestine).
ESS	Endoscopic sinus surgery	Use of endoscopic instrumentation to operate on the nose.
EUA	Exam under anesthesia	The use of anesthesia to conduct a surgical examination of a sensitive part of the body such as the eye or rectum.
I&D	Incision and drainage	Incision into and removal (drainage) of pus/fluid from an abscess, boil or infected wound/area of the body.
ICD	Implantable cardioverter-defibrillator	Insertion/implantation of a battery-powered device that can deliver a jolt of electricity to treat cardiac arrhythmia (dangerous irregular heartbeats, frequently performed in cardiac catheterization labs).

Abbreviation	Surgical Procedure	Meaning
IMN	Intramedullary nail	Insertion/implantation of a nail or rod, into the medullary cavity (center) of a long bone, such as the femur or tibia.
IOL	Intraocular lens	Insertion/implantation of a lens within the eye to treat cataracts (clouding of the lens of the eye) or myopia (nearsightedness).
IORT	Intraoperative radiation therapy	Use of therapeutic levels of radiation to treat exposed cancer tumors during surgery, frequently done during breast surgery.
LAVH	Laparoscopic-assisted vaginal hysterectomy	A visualization and treatment of pelvic organs (laparoscopy) followed by removal of the uterus through the vagina.
L&B	Laryngoscopy and bronchoscopy	A visual examination of the larynx and the bronchi.
LP	Lumbar puncture	A collection of cerebrospinal fluid (CSF) for diagnostic or therapeutic purposes. Commonly referred to as a spinal tap (not a surgical procedure, but frequently done in surgery for children).
MIDCAB	Minimally invasive direct coronary artery bypass	Use of a small incision in the ribs (mini-thoracotomy) to access the heart to bypass diseased coronary arteries. This is done “off-pump” without the assistance of the heart-lung machine.
MIS	Minimally invasive surgery	A type of surgery that uses endoscopic techniques and instrumentation in order to reduce trauma to the body during surgery (this technique reduces not only trauma to the patient’s body, but reduces recovery time and reduces infection risks, as well).
ORIF	Open reduction internal fixation	Treatment of a fractured bone with an incision and the use of plates and screws or pins to hold the fragments together.
PAL	Power assisted liposuction	Aspiration (suction) of fat cells by means of a motorized hand piece and cannula (tube).
PCI	Percutaneous coronary intervention	Coronary angioplasty; a procedure to open narrowed coronary arteries, normally performed in cardiac catheterization labs.
PDA	Patent ductus arteriosus	Repair of a congenital heart disorder in a neonate (a child less than one month old).
PEG	Percutaneous endoscopic gastrostomy	Insertion of a feeding tube (PEG tube) into the stomach through the abdominal wall.
PICC	Insertion of peripherally inserted central catheter	An intravenous catheter inserted in a peripheral vein (such as the arm) for long-term IV access (a sterile procedure, not a surgery).
PLIF	Posterior lumbar interbody fusion	Spinal surgery in which bone grafts/implants are used to fuse vertebrae in the lumbar spine—approached from the back side of the patient’s body.

Abbreviation	Surgical Procedure	Meaning
STSG	Split thickness skin graft	Healthy skin is taken from a place on the body, called the donor site, and used as a skin graft elsewhere on the body.
TAH	Total abdominal hysterectomy	Surgical removal of the uterus through an incision in the abdomen.
TEE	Transesophageal echocardiogram	Obtaining an ultrasound image of the heart by means of a probe/transducer inserted into the esophagus (frequently done in connection with cardiac surgery).
THA	Total hip arthroplasty	Hip joint reconstruction by removing the bone and placing a plastic/metal component in the hip socket (the acetabulum), as well as the femoral head (the proximal femur), resulting in a completely rebuilt joint.
TKA	Total knee arthroplasty	Knee joint reconstruction by placing implants on the end of the femur and the top of the tibia, as well as on the knee cap (the patella), resulting in a completely rebuilt joint.
TURP	Transurethral resection of the prostate	Surgical removal of part of the prostate gland by inserting instruments across the urethra to reach the prostate internally.
VATS	Video assisted thoracoscopic surgery	Use of endoscopic instruments to access the chest cavity/lungs/thorax.
VP Shunt	Ventriculo-peritoneal shunt	Surgical placement of a drain (shunt) to transfer excess cerebrospinal fluid (CSF) from the brain (ventricle) to the abdominal lining (peritoneum).
Wound VAC	Wound vacuum assisted closure	Application of a device/dressing that provides negative pressure (vacuum) to facilitate wound closure.
XLIF	Extreme lateral interbody fusion	Spinal fusion surgery approaching from the side of the patient using special instrumentation to reduce trauma on the patient's body.

ANATOMY OF A SURGICAL PROCEDURE

Once CS professionals understand medical terminology with its numerous prefixes, roots and suffixes, they will also be able to better understand the surgical procedures referred to in surgery schedules and the instruments and supplies that will be requested. Challenges arise when the OR is supplied with incorrect instrumentation and supplies for a surgical procedure. When the OR receives a case cart for a scheduled procedure from CS with the incorrect instrumentation and supplies, their attention is diverted away from the patient and onto getting the correct items in the room.

FROM MINIMALLY INVASIVE SURGERY TO OPEN PROCEDURES

The current practice in surgery is always to use a minimally invasive procedure when possible. Over the last 100 years, the vast majority of surgical procedures performed on patients have been “open” procedures. These procedures feature an incision to “open” the patient’s body in order to perform whatever the surgical procedure requires. For example, if a patient needed an appendectomy, an incision would be made in order to remove the appendix. Similarly, if the patient needed a cholecystectomy, an incision would be made to remove that patient’s gallbladder. Even open heart surgery has been handled in the same manner. If the patient had a diseased heart, the only way to operate on the heart was to split the patient’s sternum open in order to access the patient’s heart. This is no longer the only way to do surgery.

In today’s high-technology surgical climate, many procedures start with some kind of a minimally invasive surgery (MIS) approach (developed during the 1960-1970s). This means that instead of simply cutting the patient’s body open to access an organ or body part, some kind of special instruments will be used in order to perform the surgery. For example, some orthopedic procedures begin with something called a closed reduction (treatment of a fractured bone without an incision) prior to attempting an open reduction internal fixation

(implantation of plates/screws, etc.). If a closed reduction is not possible/successful, the surgeon might attempt a percutaneous pinning, which means that the surgeon might be able to access the broken bone through the skin (percutaneous) in order to reduce that fracture and stabilize it by insertion of a Steinmann pin/K-wire. If these approaches are unsuccessful (or not possible given the patient’s condition), the surgeon will perform an open reduction internal fixation of the fracture (plates and screws will be used internally to stabilize the fracture).

Similarly, if the patient needs their gallbladder removed, in most cases the procedure will be booked as laparoscopic versus an open cholecystectomy. This means that the surgeon will start by attempting to remove the patient’s gallbladder using endoscopic/laparoscopic instrumentation, as this technique is less traumatic to the patient’s body. The surgeon will only convert to an open procedure (using a larger incision to access the involved organ) if the laparoscopic approach is unsuccessful. This is also very common in thoracic surgery as many surgical procedures involving the chest start with what is called a thoracoscopic surgery (use of endoscopic instruments to access the lungs) and only convert to an open thoracotomy (incision into the thorax/chest) if the thoracoscopic approach is unsuccessful.

Other surgical procedures are following a less-invasive approach in an effort to reduce trauma to the patient. In the case of breast surgery, for example, treatment will start with a minimally-invasive biopsy, followed by a lumpectomy, if indicated. If a mastectomy is required (surgical removal of a breast), the procedure will be booked in a manner that reduces trauma to the patient’s body, while still effectively treating the disease. Therefore, a partial mastectomy can be performed (removal of a portion of the breast), or a radical mastectomy can be performed (complete removal of the breast). The same is also true for prostate surgery in men. If possible, the surgeon will attempt a transurethral resection of the prostate (TURP) using special instrumentation to reduce patient trauma. Only if the patient cannot be treated adequately using an endoscopic technique

will the surgeon perform an open procedure called a radical prostatectomy. In today's world of surgical technology, it is even possible to have a total joint replacement (arthroplasty) using some kind of MIS technique, such as a unicompartmental knee arthroplasty.

Not all patients are candidates for MIS. Some illnesses and injuries are not remedied by such an approach. For example, if a patient's cancer has grown significantly inside an organ or another area of the body, that cancer will need to be resected (removed) by means of an open procedure. Also, if a patient has a severely broken femur, this patient will require the introduction of a trochanteric (intramedullary) nail to stabilize that fracture (a form of open reduction internal fixation). If a patient's gallbladder is about to rupture, the surgeon might elect to perform an open cholecystectomy without attempting a laparoscopic approach.

CS technicians must understand that contemporary surgery is always attempted to be minimally invasive to the patient as these approaches cause less trauma to the patient's body, reduce pain, reduce recovery time, and shorten length of hospital stay. When reading surgery schedules, it is vital that CS professionals understand that, in many cases, a laparoscopy will precede a laparotomy, an arthroscopy will precede an arthrotomy, or a thoracoscopy will precede a thoracotomy. The CS professional must know that minimally invasive procedures utilize a different assortment of instruments and supplies than open procedures. In some cases, the OR will only require the endoscopic instrumentation and supplies. In other cases, OR personnel will require both "set ups," one for the endoscopic procedure and one for the open procedure.

As important as endoscopic procedures are in reducing patient trauma, new advances in surgical techniques have evolved, which can be even less invasive than traditional MIS. Whether it is the removal of the gallbladder (cholecystectomy), removal of the uterus (hysterectomy) or the removal of the prostate (prostatectomy), these procedures can now be done using robotic technology. Even

open heart procedures can now be done by means of robotic technology. These procedures utilize a complex blend of highly-specialized instrumentation and supplies. This is the least invasive and most technologically-advanced form of surgery in history. CS professionals have the responsibility of processing these complex devices.

PROCEDURE APPROACH AND PURPOSE

When reading surgery schedules and interacting with OR staff, CS technicians need to understand how procedure approach and purpose are identified. When a surgical procedure is referred to in a surgery schedule, the suffix attached to the primary medical term will indicate the purpose of the procedure. For example, "oscopy" will refer to visual examination of and possible treatment, "otomy" will refer to an incision into, "ectomy" will refer to removal of, and "plasty" will refer to repair/reconstruction of.

Another key element in how a procedure can be understood relates to the approach that the surgeon will use in performing the surgery. The approach or method typically comes at the beginning of the surgical procedure term. A laparoscopic cholecystectomy will remove the patient's gallbladder using laparoscopic instrumentation and techniques. A robotically-assisted prostatectomy will remove the patient's prostate by using robotic instrumentation and techniques. A bilateral myringotomy will make incisions for drainage into the patient's eardrum/tympanic membrane on both sides. An anterior cervical fusion will fuse some of the patient's cervical vertebrae approaching from the front of the patient's body. A posterior lumbar interbody fusion will fuse some of the patient's lumbar vertebrae approaching from the back of the patient. A vaginal hysterectomy will remove the patient's uterus via the vaginal canal. A total abdominal hysterectomy will remove the uterus through an open incision in the abdomen. All of these procedures utilize completely different instrument sets and supply packs.

CASE CARTS AND INSTRUMENT TRAYS

In the past, the surgical team took responsibility for “picking” the needed instrumentation and supplies for their scheduled procedures. Some facilities still function in this manner; however, in recent years it has become more common for CS professionals to perform this function. Therefore, CS technicians may be responsible for reading the OR surgery schedule in order to print pick lists, obtain the needed instrumentation, and even find/obtain instrumentation that is already “in use” in another surgical procedure.

Surgery schedules are prioritized according to patient needs, which allows the OR to schedule procedures for the next day or following week. If a case is booked as a thoracoscopy (possible thoracotomy), for example, pick lists may be

required (one for the endoscopic portion and another for the possible open procedure). **Figure 2.11** provides a basic example of a surgery schedule.

Many procedures in today’s surgical climate require what is called “loaner” instrumentation. These are sets of instruments (frequently multiple sets of instruments) that are shipped to the hospital specifically for use by a specific physician on a specific patient. These instruments may be used one time and then returned to the vendor. To keep surgical procedures on schedule, it is also essential that CS professionals anticipate the need for loaner instrumentation. This helps ensure that devices are delivered well in advance to allow for safe, effective reprocessing and prompt delivery to the OR. More information on loaner instrumentation is discussed in Chapter 11.

Sample Surgery Schedule					
Understanding Medical Terms can help CS Technicians better understand procedural needs.					
Room	Start Time	Patient	Surgeon	Procedure	Comments
1	7:00	John Doe	Dr. Wells	Cystoscopy with Left Retrogrades, Left Stent Insertion	C-Arm
	TF	T m Smith	Dr. Wells	Transurethral Resection of the Prostate	
2	7:00	Jon Williams	Dr. Green	Septoplasty, Possible Endoscopic Sinus Surgery	
	TF	Tammi Jones	Dr. Green	Tonsillectomy	
3	7:00	Mary Smith	Dr. Clark	TAH with BSO	
	TF	Jane Doe	Dr. Clark	D & C	
4	7:00	Bill Williams	Dr. Jones	TKA Right – Revision	Loaner Instruments
	TF	Sally Sims	Dr. Jones	ORIF - Left Ankle	
	TF	Bob Roberts	Dr. Jones	CR, Possible Pinning Right Thumb	Possible ORIF

Figure 2.11

THE VALUE OF MEDICAL TERMINOLOGY IN THE OPERATING ROOM

Awareness within the Operating Room

Knowledge of medical terminology and language is essential for ensuring that CS professionals have a clear understanding of what the OR and other healthcare customers require for their procedures. Prefixes, roots and suffixes discussed in this chapter are not mere items to remember or memorize. Instead, they are vital tools that will allow CS professionals to become more proficient in the department, improve communication with the OR and other healthcare customers, and provide safe, consistent, high-quality service that drives positive patient outcomes.

CONCLUSION

Medical and surgical staff members appreciate when everything goes smoothly with their patients/cases. The patient's needs are the motivating factor of every Central Service professional. CS professionals must respond to Operating Room staff in a timely, effective way. It is the responsibility of the CS department to assist the surgical and medical staff to deliver the highest quality of care to each and every patient. Communicating effectively and successfully meeting the needs of the OR and other healthcare customers requires CS professionals to possess a functional, ever-evolving understanding of medical terminology and language.

RESOURCES

Association for the Advancement of Medical Instrumentation. ANSI/AAMI ST79, A4 2013, Section 2. *Comprehensive Guide to Steam Sterilization and Sterility Assurance in Health Care Facilities*.

The Joint Commission. *Hospital Accreditation Standards, IC-10*. 2014.

Centers for Disease Control and Prevention. *Guideline for Disinfection and Sterilization in Healthcare Facilities*. 2008.

Leiken JB, Lipsky MS, Eds. *American Medical Association Complete Medical Encyclopedia*. Random House. 2003.

Stedman TL. *Stedman's Medical Dictionary 28th Ed.*, pp. 1047-1048. Lippincott Williams & Wilkins. 2005.

CENTRAL SERVICE TERMS

Case cart

STAT

Word elements

Root word element

Prefix word element

Suffix word element

Combining vowel